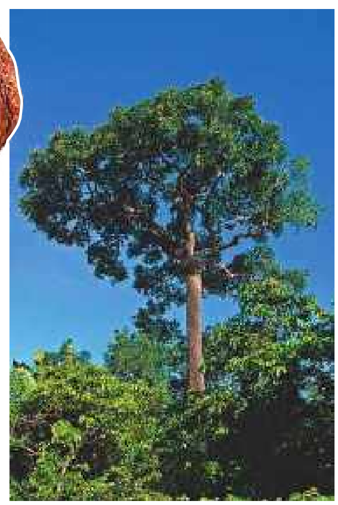
▼ Figure 53.14 Variation in the number and size of seeds in plants.

(a) Dandelions grow quickly and release a large number of tiny fruits, each containing a single seed. Producing numerous seeds increases the chance that at least some will grow into plants that eventually produce seeds themselves.



(b) Some plants, such as the Brazil nut tree (right), produce a moderate number of large seeds in pods (above). Each seed's large endosperm provides nutrients for the embryo, an adaptation that helps a relatively large fraction of offspring survive.

distances to a broader range of habitats **(Figure 53.14a)**. Animals that suffer high predation rates, such as quail, sardines, and mice, also tend to produce many offspring.

In other organisms, extra investment on the part of the parent greatly increases the offspring's chances of survival. Brazil nut and walnut trees produce large seeds packed with nutrients that help the seedlings become established **(Figure 53.14b)**. Primates generally bear only one or two offspring at a time; parental care and an extended period of learning in the first several years of life are very important to offspring fitness. Such provisioning and extra care can be especially important in habitats with high population densities.

One way to categorize variation in life history traits is related to the logistic growth model discussed in Concept 53.3. Selection for traits that are advantageous at high densities is referred to as ***K*-selection**. In contrast, selection for traits that maximize reproductive success in uncrowded environments (low densities) is called ***r*-selection**. These names follow from the variables of the logistic equation. *K*-selection is said to operate in populations living at a density near the limit imposed by their resources (the carrying capacity, *K*), where competition among individuals is stronger. Mature trees growing in an old-growth forest are an example of *K*-selected organisms. In contrast, *r*-selection is said to maximize *r*, the intrinsic rate of increase, and occurs in environments in which population densities are well below carrying capacity or individuals face little competition. Such conditions are often found in disturbed habitats that are being recolonized. Weeds growing in an abandoned agricultural field are an example of *r*-selected organisms.

The concepts of *K*- and *r*-selection represent two extremes in a range of actual life histories. The framework of *K*- and *r*-selection, grounded in the idea of carrying capacity, also relates to the important question we alluded to earlier: *Why* does population growth rate decrease as population size approaches carrying capacity? Answering this question is the focus of the next section.

CONCEPT CHECK 53.4

1. Identify three key life history traits, and give examples of organisms that vary widely in each of these traits.
2. In the fish called the peacock wrasse (*Symphodus tinca*), females disperse some of their eggs widely and lay other eggs in a nest. Only the latter receive parental care. Explain the trade-offs in reproduction that this behavior illustrates.
3. **WHAT IF?** Mice that experience stress such as a food shortage will sometimes abandon their young. Explain how this behavior might have evolved in the context of reproductive trade-offs and life history.

For suggested answers, see Appendix A.

CONCEPT 53.5

Density-dependent factors regulate population growth

What environmental factors keep populations from growing indefinitely? Why are some populations fairly stable in size, while others are not?

Answers to such questions can be important in practical applications. Farmers may want to reduce the abundance of insect pests or stop the growth of a weed that is spreading rapidly. Conservation ecologists need to know what environmental factors create favorable feeding or breeding habitats for endangered species, such as the white rhinoceros and the whooping crane. Overall, whether seeking to reduce the size of an unwanted population or increase the size of one that is endangered, it is helpful to understand factors that affect population abundance.

Population Change and Population Density

To understand why a population stops growing when it reaches a certain size, ecologists study how the rates of birth, death, immigration, and emigration change as population density rises. If immigration and emigration offset